Possibilistic Variational Inference

Learning the Precision of the Gaussian Variational Candidate

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Internship Context

- April to August 2026, jointly at
  - LPSM, CNRS and Sorbonne Université, Paris, 1.5 months (background phase)
  - SPMS, Nanyang Technological University, Singapore, 3 months (reproduction, landscapes, contribution & report writing)
- Supervised by Badr-Eddine Chérief-Abdellatif (LPSM) and Jérémie Houssineau (NTU)

The paper this internship extends

J. Singh, S. Wongso, J. Houssineau, B.-E. Chérief-Abdellatif. *Maxitive Donsker–Varadhan Formulation for Possibilistic Variational Inference*. arXiv:2511.21223, 2026 [1].

What this internship addresses

Singh et al. [1]: **expectation** learned, but **precision** fixed by hand or tuned heuristically, never derived from the variational objectives themselves.

Outline

Possibilistic Variational Inference and Its Limits

Learning the Precision of a Gaussian Possibilistic Candidate

SAM Application, Algorithms, and Experimental Validation

Discussion and Conclusion

Possibilistic Variational Inference and Its Limits

Epistemic Uncertainty and Probability Dilution

- **Epistemic uncertainty**: lack of knowledge about a fixed, deterministic $\theta^\star$, with no repeatable experiment to draw from.
- **Improper priors**: $\pi(\theta) \propto 1$ on unbounded Θ is not integrable, and Jeffreys' prior $\pi_J \propto \sqrt{\det I(\theta)}$ can be improper.
- **False confidence** [2]: as $\sigma \to \infty$, $\mathbb{P}(\theta^\star \in A) \to 0$ even when $\mu \in A$ always.
- **Key intuition**: possibility theory [3, 4], a framework representing complete ignorance without diluting local plausibility.

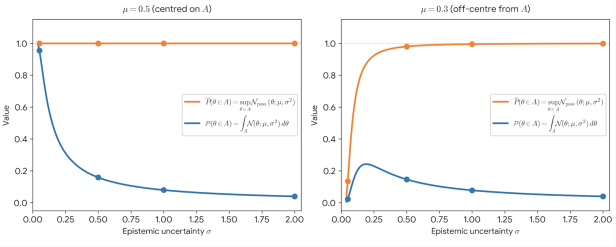


Figure. Probability dilution for a target set $A = [0.4, 0.6]$, at $\mu = 0.5$ and $\mu = 0.3$.

Possibility Functions and Possibilistic Inference

Definition (Possibility function, 5)

$f : \Theta \to [0, 1]$ measurable with $\sup_\theta f(\theta) = 1$. Associated possibility measure: $\bar{P}(A) \coloneqq \sup_{\theta \in A} f(\theta)$. Candidate space $\mathcal{F}(\Theta)$ ($\sup g = 1$) replaces $\mathcal{P}(\Theta)$ ($\int q = 1$), uninformative prior $\pi \equiv 1$ naturally proper.

Possibilistic Bayes' rule, 5

$$g^\star_{\max}(\theta) = \frac{e^{-\ell(\theta)}\pi(\theta)}{Z_{\max}} = \frac{e^{-\hat{\ell}(\theta)}}{Z_{\max}}, \qquad Z_{\max} \coloneqq \sup_{\theta \in \Theta} e^{-\hat{\ell}(\theta)}$$

$\hat{\ell} \coloneqq \ell - \log \pi$. $Z_{\max}$ measures **consistency** between prior and likelihood, not average marginal fit.

Definition (Possibilistic expectation, 5)

When unique, the expectation of $f \in \mathcal{F}(\Theta)$ is its mode, $\mathbb{E}^*[\theta]_f \coloneqq \arg\max_{\theta \in \Theta} f(\theta)$, not an integral mean.

- **Practical**: a single deterministic pass at the mode suffices at test time, no Monte Carlo sampling.
- **Structural**: $Z_{\max}$ already an optimization (sup), turning VI from integral-to-optimization into optimization-to-optimization.

The Maxitive Donsker–Varadhan Formula

Theorem (Maxitive Donsker–Varadhan [1])

Let $\pi : \Theta \to (0, 1]$ and $h \coloneqq -\ell$. Then:

$$\log Z_{\max} \coloneqq \log \sup_{\theta \in \Theta} e^{h(\theta)} \pi(\theta) = \sup_{g \in \mathcal{F}(\Theta)} \inf_{\theta} \left\{ h(\theta) - \log \tfrac{g(\theta)}{\pi(\theta)} \right\} = \inf_{g \in \mathcal{F}(\Theta)} \sup_{\theta} \left\{ h(\theta) - \log \tfrac{g(\theta)}{\pi(\theta)} \right\}$$

- **Probabilistic Donsker–Varadhan analogue** [6] (ν: reference measure, $\int e^h \, \mathrm{d}\nu < \infty$):

  $$\log \int_{\Theta} e^{h(\theta)} \, \nu(\mathrm{d}\theta) = \sup_{\rho \in \mathcal{P}(\Theta)} \left\{ \int h \, \mathrm{d}\rho - \mathrm{KL}(\rho \| \nu) \right\}$$

  recovers the ELBO at $h = -\ell$, $\nu = \pi$: $\mathrm{ELBO}(q) \coloneqq -\mathbb{E}_q[\ell(\theta)] - \mathrm{KL}(q \| \pi)$.
  $\mathrm{KL}(q \| p) \coloneqq \int \log \frac{\mathrm{d}q}{\mathrm{d}p} \, \mathrm{d}q$ if $q \ll p$, $+\infty$ otherwise.

- **Two variational objectives vs. one single ELBO**:
  - Sup-inf yields the **lower bound**: $\underline{\mathrm{CBO}}(g) \coloneqq \inf_\theta \left\{ -\ell(\theta) - \log \frac{g(\theta)}{\pi(\theta)} \right\}$
  - Inf-sup yields the **upper bound**: $\overline{\mathrm{CBO}}(g) \coloneqq \sup_\theta \left\{ -\ell(\theta) - \log \frac{g(\theta)}{\pi(\theta)} \right\}$

LCBO, UCBO, and the Sandwich Decomposition

Exact sandwich identity:

$$\underline{\mathrm{CBO}}(g) + D_{\max}(g \| g^\star_{\max}) = \log Z_{\max} = \overline{\mathrm{CBO}}(g) - D_{\max}(g^\star_{\max} \| g)$$

$$\Longrightarrow \ \underline{\mathrm{CBO}}(g) \le \log Z_{\max} \le \overline{\mathrm{CBO}}(g)$$

Probabilistic analogue (ELBO):

$$\log Z_{\mathrm{add}} = \mathrm{ELBO}(q) + \mathrm{KL}\big(q \,\|\, \pi(\cdot \mid x_{1:n})\big)$$

Max-relative entropy ($D_{\max} \ge 0$, $+\infty$ if $\mathrm{supp}(g) \not\subseteq \mathrm{supp}(f)$):

$$D_{\max}(g \| f) := \sup_{\theta \in \mathrm{supp}(f)} \log \frac{g(\theta)}{f(\theta)}$$

- **Maximising $\underline{\mathbf{CBO}}$**: mode-seeking (under-estimates $g^\star$), $\arg\max_g \underline{\mathrm{CBO}}(g) = \{g \in \mathcal{F} : g \preceq g^\star_{\max}\}$.
- **Minimising $\overline{\mathbf{CBO}}$**: support-covering (over-estimates $g^\star$), $\arg\min_g \overline{\mathrm{CBO}}(g) = \{g \in \mathcal{F} : g^\star_{\max} \preceq g\}$.
- **Intersection**: $g^\star_{\max}$ the unique canonical solution, $\{g^\star_{\max}\} = \arg\max_g \underline{\mathrm{CBO}}(g) \cap \arg\min_g \overline{\mathrm{CBO}}(g)$.

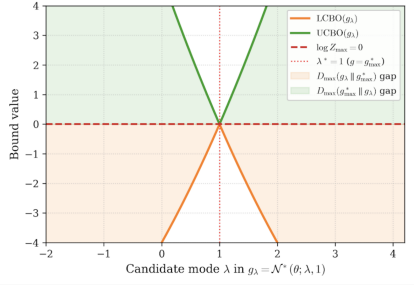


Figure. Sandwich decomposition: lower and upper bounds sandwiching $\log Z_{\max}$.

The CBOpt Updates and the Gap They Leave Open

Signed objective and update [1]

$$\mathrm{CBO}_s(g) := \operatorname*{ext}_{\theta s}\Big\{-\ell(\theta) - \log \tfrac{g(\theta)}{\pi(\theta)}\Big\}, \qquad \eta_{t+1} = \eta_t - \rho_t\, H_s^{-1}\nabla\hat{\ell}(\eta_t) = \eta_t - \rho_t\, s\, \frac{\bar{g}_t + \delta\eta_t}{h_t + \delta - \vartheta_t}$$

$s \in \{+1, -1\}$. Signed curvature $H_s \approx s\ \mathrm{diag}(h_t + \delta - \vartheta_t) \succ 0$. $\bar{g}_t, h_t$: mini-batch gradient and diagonal curvature. δ: L^2 prior precision, division element-wise.

- **Candidate**: restricted to isotropic Gaussian $\mathcal{N}^*(\theta; \eta, \vartheta^{-1} I)$.
- **Curvature tracking**: ensures $H_s \succ 0$ via $\vartheta_t = \gamma c_t$ with scaling factor γ and moving envelope c_t (momentum β_3, shifted curvature $\tilde{h}_t := h_t + \delta$):
  - $s = +1$: $\vartheta_t < \tilde{h}_t \implies \gamma \in [0, 1)$, $c_t = \min\{\beta_3 c_{t-1}, \tilde{h}_t\}$ (`uCBOpt-adapt`)
  - $s = -1$: $\vartheta_t > \tilde{h}_t \implies \gamma > 1$, $c_t = \max\{\beta_3 c_{t-1}, \tilde{h}_t\}$ (`lCBOpt-adapt`)

The gap

In neither case is ϑ derived or optimised from the variational objective itself.

Landscapes on Tractable Posteriors

Four tractable targets with known $g^{\star}_{\max}$ in closed form: Gaussian (sanity check), Bernoulli–Beta, Laplace, bimodal mixture.

Systematic pattern

Both $D_{\max}$ terms collapse to numerical noise wherever the candidate family matches the target's structure. Whenever there is a structural mismatch (support, tails, multimodality), one direction explodes by orders of magnitude.

Order-p check

A softened $D_p \to D_{\max}$ test confirms the ranking is invariant, gaps stem from family misspecification, not from $D_{\max}$ itself.

Left unexplained

Directly maximising $\underline{\mathrm{CBO}}$ over precision collapses ($\vartheta \to \infty$), the phenomenon formalised next.

Learning the Precision of a Gaussian Possibilistic Candidate

Formalising the Collapse

Isotropic candidate $g_{\eta,\vartheta}$, L^2 prior of precision δ, ℓ twice continuously differentiable (almost everywhere for ReLU networks [7]). Now seek to update ϑ at fixed η, letting ϑ vary freely.

Lemma (Degeneracy of the raw objective)

$\vartheta \mapsto \mathrm{CBO}_s(\eta, \vartheta)$ is strictly increasing on its admissible domain for both signs. The optimum sits only at the boundary ($\vartheta \to 0^+$ for $\overline{\mathrm{CBO}}$, $\vartheta \to \infty$ for $\underline{\mathrm{CBO}}$).

Idea

Proof sketch: by Danskin's theorem, with inner extremiser $\theta_s := \mathrm{argext}_s\{-\hat{\ell}(\theta) + \frac{\vartheta}{2}\|\theta - \eta\|^2\}$ and Newton step $u_s := -H_s^{-1}\nabla\hat{\ell}(\eta)$ ($\theta_s - \eta \approx s\, u_s$): $\partial_\vartheta \mathrm{CBO}_s(\eta, \vartheta) = \frac{1}{2}\|u_s\|^2 > 0$ ($u_s \neq 0$ away from stationarity). No interior stationary point can exist.

Exactly the collapse observed when fitting the Gaussian candidate in the landscape analysis: the raw CBO alone cannot identify an interior precision.

A Geometric Penalty via Level-Set Log-Volume

Definition (Level-set volume)

For $g_{\eta,S} = \mathcal{N}^*(\theta; \eta, S^{-1})$, the level set $\{g_{\eta,S} \geq \alpha\}$ is an ellipsoid of volume

$$\mathrm{vol}_\alpha(S) = V_d(-2\log\alpha)^{d/2}(\det S)^{-1/2} \implies \log \mathrm{vol}_\alpha(S) = -\tfrac{1}{2}\log\det S + c(\alpha, d)$$

- **No normaliser in possibility theory**: $\sup g = 1$, no natural entropy penalises concentration
- **Geometric cost**: penalise by the log-volume of the candidate's level sets *(suggested by S. Wongso)*, a natural proxy for informativeness, optimisation direction depending on which bound is being enforced
- **Invariance**: $c(\alpha, d)$ does not depend on S, choice of $\alpha \in (0, 1)$ irrelevant to optimization

The Penalty Selects an Extremal-Volume Candidate

Theorem (Extremal-volume selection)

At multiplier $\mu = 1$ (up to additive constants), the Lagrangian of:

(a) $\min_{\eta,S} \log \mathrm{vol}_\alpha(S)$ *s.t.* $\overline{\mathrm{CBO}}(g_{\eta,S}) \leq \log Z_{\max}$ *($g \geq g^\star$) is* PCBO_s *($s = +1$).*

(b) *For a target level $c \leq \log Z_{\max}$:* $\max_{\eta,S} \log \mathrm{vol}_\alpha(S)$ *s.t.* $\underline{\mathrm{CBO}}(g_{\eta,S}) \geq c$ *($g \leq g^\star$ at $c = \log Z_{\max}$) is* PCBO_s *($s = -1$).*

Idea

Proof sketch: form the Lagrangian $L = \log \mathrm{vol}_\alpha(S) + \mu(\mathrm{CBO} - \mathrm{cst})$, inject $\log \mathrm{vol}_\alpha(S) = -\frac{1}{2} \log\det S + c$, evaluate at $\mu = 1$.

Geometric meaning

Prevents trivial solutions: $g \geq g^\star$ trivial for 1_Θ ($\mathrm{vol} = \infty$), $g \leq g^\star$ trivial for a Dirac spike ($\mathrm{vol} = 0$). The penalty isolates the most informative non-trivial candidate.

The Penalty Repairs the Degeneracy

Proposition (Penalty breaks monotonicity)

For any $\zeta > 0$ (penalty coefficient, generalising $\mu = 1$), $\mathrm{PCBO}_s(\eta, \cdot)$ has an interior stationary point: a unique global minimum for $s = +1$ ($\overline{\mathrm{CBO}}$), a unique global maximum for $s = -1$ ($\underline{\mathrm{CBO}}$), in the exact-quadratic case. Unlike the raw objective, $\partial_\vartheta \mathrm{PCBO}_s$ now diverges at both ends of the admissible domain.

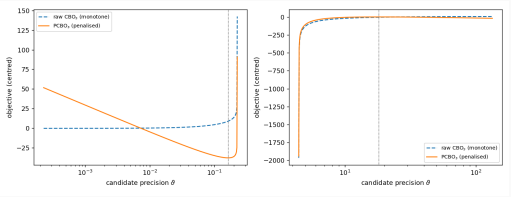


Figure. Raw versus penalised objective on the isotropic toy problem, far regime.

Four precision structures: isotropic ($S = \vartheta I$), diagonal ($S = \mathrm{diag}(\boldsymbol{\vartheta})$), low-rank-plus-diagonal (LRD), block-diagonal (BD).

Structure	Penalty $-\frac{1}{2} \log \det S$
Isotropic	$-\frac{d}{2} \log \vartheta$
Diagonal	$-\frac{1}{2} \sum_{j=1}^{d} \log \vartheta_j$
LRD	$\frac{1}{2}(d-1) \log \psi + \frac{1}{2} \log(\psi + \lVert u_s^{\mathrm{LRD}} \rVert^2)$
BD	$\sum_\ell \left(\frac{1}{2}(d_\ell - 1) \log \psi_\ell + \frac{1}{2} \log(\psi_\ell + \lVert u_{s,\ell}^{\mathrm{BD}} \rVert^2) \right)$

Table. Specificity penalty for each precision structure.

Isotropic and Diagonal Stationarity

Isotropic precision $\vartheta \in \mathbb{R}_{>0}$, diagonal precision $\boldsymbol{\vartheta} \in \mathbb{R}^d_{>0}$. Setting $\partial_\vartheta \mathrm{PCBO}_s = 0$ (isotropic) or $\partial_{\vartheta_j} \mathrm{PCBO}_s = 0$ (diagonal) gives the stationarity condition:

Proposition (Optimal precision from the Newton step)

$$\vartheta^\star \approx \frac{d}{\|u_s\|^2}, \qquad \vartheta_j^\star \approx \frac{1}{(u_s^{\mathrm{diag}})_j^2}, \qquad u_s := -H_s^{-1} \nabla \hat{\ell}(\eta)$$

This condition is **implicit** (H_s depends on ϑ), solved by damped Picard iteration (Krasnoselskii–Mann relaxation [8]), written in isotropic (scalar) form. The diagonal case applies the same map coordinatewise:

Damped Picard map, with anti-collapse/positivity cap

$$\vartheta_{t+1} = T_w(\vartheta_t) := (1-w)\vartheta_t + w\, \mathcal{C}_s\big(\Phi(\vartheta_t)\big), \qquad \Phi(\vartheta) := \frac{d}{\|u_s(\vartheta)\|^2}$$

$$\mathcal{C}_{+1}(\vartheta) := \max\big(\min(\vartheta,\ m/r,\ m(1-\epsilon_{\mathrm{rel}})),\ v_{\min}\big), \quad \mathcal{C}_{-1}(\vartheta) := \max\big(\min(\vartheta,\ M/r),\ M(1+\epsilon_{\mathrm{rel}})\big)$$

$m = \min_j(h_j + \delta)$, $M = \max_j(h_j + \delta)$: curvature bounds. r: anti-collapse divisor. ϵ_{rel}: positivity margin. $v_{\min}$: floor.

$\mathcal{C}_s$ enforces $H_s \succ 0$ and prevents Dirac collapse, keeping ϑ inside an invariant interval, a known device in natural-gradient VI [9].

Linear Convergence of the Damped Picard Map

Same scalar form as before, extended coordinatewise to the diagonal case. The caps confine the iterates to $\mathcal{I}_{+1} \coloneqq \left[v_{\min},\ m(1-\epsilon_{\mathrm{rel}})\right]$ and $\mathcal{I}_{-1} \coloneqq \left[M(1+\epsilon_{\mathrm{rel}}),\ M/r\right]$.

Theorem (Picard convergence)

Let $|\Phi'(\vartheta)| \le L_\Phi$ on an invariant interval $\mathcal{I}_s$, $\kappa(w) \coloneqq 1 + w(L_\Phi - 1)$. If $L_\Phi < 1$, T_w is a $\kappa(w)$-contraction for every $w \in (0,1]$. By Banach's fixed-point theorem it admits a unique fixed point $\vartheta^\dagger \in \mathcal{I}_s$, with $|\vartheta_t - \vartheta^\dagger| \le \kappa(w)^t |\vartheta_0 - \vartheta^\dagger|$.

Idea

Proof sketch: the cap $\mathcal{C}_s$ is a projection onto $\mathcal{I}_s$, hence non-expansive. The mean value theorem bounds Φ's Lipschitz constant by L_Φ on $\mathcal{I}_s$.

Quantity evaluated in the experiments

ϑ_0 initialised symmetrically close to the boundary, K damped Picard steps per mode update η_t. In practice, L_Φ estimated by the explicit bound $L_\Phi = 2d\|g\|^2/(u_{\min}(\eta)^4 a_{\min}^3)$, $g \coloneqq \nabla\hat{\ell}(\eta)$, $a_{\min} \coloneqq \epsilon_{\mathrm{rel}} \min_j(h_j + \delta)$, $u_{\min}(\eta) \coloneqq \min_{\mathcal{I}_s} \|u_s\| > 0$, degrading near stationary points.

Full and Block-Diagonal Precisions

Beyond isotropic and diagonal: a general precision $S \in \mathcal{S}^d_{++}$, block-diagonal $S = \bigoplus_\ell S_\ell$.

Proposition (Rank-one singularity)

Exact stationarity gives $S_\star^{-1} = u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top$, rank one and singular for $d > 1$. Regularise to a low-rank-plus-diagonal (LRD) form, similar to low-rank covariance approximations in VI [10]:

$$S^{-1} = u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top + \psi I, \qquad \psi > 0$$

ψ must certify $H_s \succ 0$ [9], signs behaving differently (BD applies the same per block, with $m_\ell, M_\ell, u_{s,\ell}^{\mathrm{BD}}$):

$s = +1$, unconditional

$\lambda_{\max}(S) < m \iff \psi > 1/m$: set $\psi = 1/m + a$ ($a > 0$: margin above $1/m$). Sufficient for any curvature.

$s = -1$, conditional

Feasible only if $\|u_s^{\mathrm{LRD}}\|^2 < 1/M$: set $\psi = \frac{1}{\varphi}\left(\frac{1}{M} - \|u_s^{\mathrm{LRD}}\|^2\right)$ ($\varphi > 1$: multiplicative damping factor). Fails early in training.

When infeasible (LRD: globally, BD: per block), flip to $s_{\mathrm{e}} = +1$ until $\|u_s^{\mathrm{LRD}}\|^2$ drops below $(1 - \xi)/M$ ($\xi \in (0, 1)$: margin avoiding a near-zero ψ), then switch back to $s_{\mathrm{e}} = -1$: the standard warm-up principle of starting broad before tightening toward the target [11].

A Chain of Results

Common to all four structures

Raw CBO degenerate $\rightarrow$ log-volume penalty selects extremal-volume candidate

$\downarrow$

Isotropic / Diagonal

Implicit fixed point $\rightarrow$ damped Picard step under caps (linear convergence on $\mathcal{I}_s$) $\rightarrow$ `CBOpt-iso` / `CBOpt-diag`

LRD / BD

Stationarity singular $\rightarrow$ LRD regularisation $\rightarrow$ sign-asymmetric ψ safeguards (unconditional for $s = +1$, warm-up flip for $s = -1$) $\rightarrow$ `CBOpt-LRD` / `CBOpt-BD`

SAM Application, Algorithms, and Experimental Validation

Lower CBO and Local Robustness: Connecting to SAM

An application of the framework above, first derived by Shelvia Wongso [12], connecting to a classical robust-optimisation setting.

Sharpness-Aware Minimization (SAM) [13] minimizes local worst-case loss (here, $\delta = 0$):

$$\mathcal{E}_{\mathrm{SAM}}(\eta;\rho) \coloneqq \sup_{\|\varepsilon\|_2 \le \rho} \ell(\eta+\varepsilon)$$

Proposition (Indicator candidate is SAM, 12)

For $\Theta \subseteq \mathbb{R}^d$ closed, ℓ continuous, flat prior $\pi \equiv 1$, and $g_{\eta,\rho}(\theta) \coloneqq 1[\|\theta-\eta\|_2 \le \rho]$: $-\log g_{\eta,\rho} = +\infty$ outside the ball forces the $\underline{\mathrm{CBO}}$ ($s=-1$) infimum to match worst-case loss.

$$\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\sup_{\|\theta-\eta\|_2 \le \rho} \ell(\theta) \implies \arg\max_{\eta} \underline{\mathrm{CBO}}(g_{\eta,\rho}) = \arg\min_{\eta} \mathcal{E}_{\mathrm{SAM}}(\eta;\rho)$$

If $\nabla\ell(\eta) \neq 0$ and $B(\eta,\rho) \subseteq \Theta$, the first-order step is $\varepsilon^\star_{\mathrm{SAM}} = \rho\,\nabla\ell(\eta)/\|\nabla\ell(\eta)\|_2$.

Exact identity of objectives, not just stationary points. Unlike the probabilistic SAM–Bayes link, which only holds as a convex relaxation [14].

Gaussian Relaxation and the Adaptive Radius

Limitation: ρ is fixed manually in SAM, motivating a Gaussian candidate with a learned radius.

Proposition (Gaussian relaxation of SAM, 12)

For $g_{\eta,\vartheta} = \mathcal{N}^*(\cdot; \eta, \vartheta^{-1} I)$, the lower CBO relaxes the hard ball into a quadratic penalty:

$$\underline{\mathrm{CBO}}(g_{\eta,\vartheta}) = -\sup_{\theta \in \mathbb{R}^d} \left\{ \ell(\theta) - \tfrac{\vartheta}{2} \|\theta - \eta\|_2^2 \right\}$$

For $L < \vartheta$ (L: Lipschitz constant of $\nabla \ell$), the exact maximiser $u^\star$ exists and is unique. Its first-order surrogate is $u^\star_{\mathrm{soft}} = \vartheta^{-1} \nabla \ell(\eta)$, exact when ℓ is affine near η.

Superlevel-set bridge: $\{g_{\eta,\vartheta} \geq \alpha\} = \bar{B}(\eta, \rho_\alpha)$, $\rho_\alpha := \vartheta^{-1/2} \sqrt{-2 \log \alpha}$, giving $\rho_\alpha = \vartheta^{-1/2}$ at $\alpha = e^{-1/2}$.

Adaptive SAM radius derived from the learned precision

With $\vartheta^\star = d / \|u_{-1}\|_2^2$ and $u_{-1} \approx -H_{-1}^{-1} \nabla \hat{\ell}(\eta)$:

$$\rho^\star(\eta) \approx \frac{1}{\sqrt{d}} \left\| H_{-1}^{-1} \nabla \hat{\ell}(\eta) \right\|_2$$

- **Direction vs norm**: $u^\star_{\mathrm{soft}}$ shares SAM's direction, but its norm scales as $\vartheta^{-1} \|\nabla \ell(\eta)\|$, vs SAM's fixed ρ
- **Hyperparameter trade-off**: replaces direct tuning of ρ with the spread safeguards $(r, \epsilon_{\mathrm{rel}})$

Algorithms and Complexity

`uCBOpt`/`lCBOpt` $\times$ {iso, diag, LRD, BD} ($s = \pm 1$) differ in storage and inversion cost:

Structure	Storage	$H_s^{-1}(\bar{g}_t + \delta\eta_t)$
Isotropic	$O(1)$	$O(d)$
Diagonal	$O(d)$	$O(d)$
LRD	$O(d)$	$O(d)$ (Sherman–Morrison)
Block-diagonal	$O(d)$	$O(d)$

Table. Per-mode-step storage and inversion cost of each precision structure.

- **Predicted order**: $\mathrm{LRD} \approx \mathrm{BD} < \mathrm{iso} \approx \mathrm{diag}$, the closed-form pair ($\psi$ in closed form, no iteration) against the K-iteration pair ($O(Kd)$)
- **LRD inversion**: $S^{-1} = uu^\top + \psi I$ stored as $(u, \psi) \in \mathbb{R}^d \times \mathbb{R}_{>0}$, never as a dense $d \times d$ matrix. Sherman–Morrison solves $H_s^{-1} z$ in $O(d)$, avoiding the $O(d^3)$ cost of a naive inversion

Toy Problem, Sanity Checks and Genuine Findings

Exact quadratic loss, $d = 30$, closed form, no minibatch or EMA noise.

Sanity checks (hold by construction)

- Regularity (assumption in the report): $\theta_s(\vartheta)$ finite and unique across the admissible grid, both signs
- Monotonicity and its repair: raw CBO strictly monotone, penalty gives an interior stationary point (already shown)
- SAM direction: $\varepsilon^\star_{\mathrm{SAM}}$ and $u^\star_{\mathrm{soft}}$ collinear to machine precision ($\cos = 1.000000$)

Genuine findings

- Penalty coefficient sensitivity: $\vartheta^\star$ moves 2.8× ($s = +1$, sub-linear) and 6.4× ($s = -1$, more sensitive) over a 16× coefficient range, justifying $\zeta = d/2$ rather than a hand-picked value
- Dimension sweep ($d \in [3.5 \times 10^3, 1.9 \times 10^5]$): confirms the predicted pairing, refined to $\mathrm{LRD} < \mathrm{BD} < \mathrm{iso} < \mathrm{diag}$ and measured factor $\sim 2\times$ rather than K=5×, likely GPU overhead

Reproduction and Benchmark on LeNet-5 and Fashion-MNIST

LeNet-5 ($d = 44{,}426$), Fashion-MNIST (in-domain) vs EMNIST-Letters (OOD), 100 epochs, 3 seeds.

Optimiser	Acc. (%) ↑	NLL ↓	ECE ↓	FPR@95 ↓	AUROC ↑
`uCBOpt-iso` (learned ϑ)	91.2 ± 0.1	0.257 ± 0.002	0.022 ± 0.001	$\mathbf{0.263 \pm 0.015}$	$\mathbf{0.793 \pm 0.018}$
`uCBOpt-diag` (learned $\boldsymbol{\vartheta}$)	$\mathbf{91.5 \pm 0.1}$	$\mathbf{0.252 \pm 0.001}$	0.020 ± 0.002	0.292 ± 0.018	0.758 ± 0.018
`uCBOpt-BD` (learned S)†	90.8 ± 0.2	0.256 ± 0.001	$\mathbf{0.014 \pm 0.001}$	0.289 ± 0.017	0.755 ± 0.018
fixed `uCBOpt` [1]	90.9 ± 0.2	0.272 ± 0.004	0.025 ± 0.001	0.263 ± 0.005	0.795 ± 0.005
IVON (64 MC samples) [15]	$\mathbf{91.0 \pm 0.2}$	$\mathbf{0.258 \pm 0.003}$	$\mathbf{0.019 \pm 0.001}$	$\mathbf{0.241 \pm 0.009}$	$\mathbf{0.812 \pm 0.012}$
AdamW [16]	89.6 ± 0.3	0.296 ± 0.005	0.022 ± 0.006	0.335 ± 0.005	0.722 ± 0.016

Table. Reproduction and learned-precision benchmark. All `CBOpt` variants use a single deterministic pass.
† *Seeds 0, 1 (seed 2 diverged at epoch 16).*

- **Reproduction**: all reference baselines matched within 2σ of Singh et al. [1]
- **In-domain gains**: learned precision improves Acc, NLL, and calibration, from a **single pass** (64× cheaper)
- **OOD trade-off**: the three extensions score below fixed `uCBOpt` and IVON on FPR@95/AUROC. Wider seed variance on OOD affects every precision-learning method, IVON included, matching the underspecification pattern [17]

Geometry and Safeguards

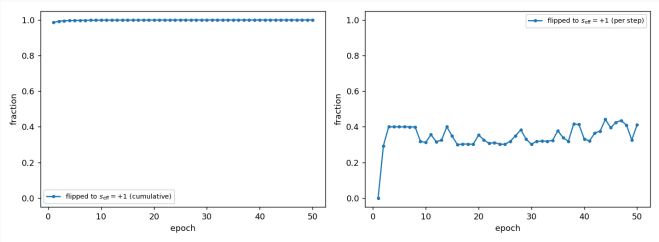


Figure. Fraction of steps or blocks flipped to $s_{\text{eff}} = +1$ for `lCBOpt-LRD` and `lCBOpt-BD`.

- $s = +1$, **iso/diag**: $H_s \succ 0$ before capping held on 0.0% of steps. The cap is the mechanism, capping ϑ to $(1 - \epsilon_{\text{rel}})(h_j + \delta)$, the same form as `uCBOpt-adapt`'s hand-designed γc_t envelope [1], matching it numerically
- $s = -1$, **LRD**: the global feasibility condition ($\|u\|^2 < 1/M$ across all d coordinates at once) never holds at network scale. `lCBOpt-LRD` flips on 100% of steps
- $s = -1$, **BD**: per-block feasibility is far easier to satisfy, only 34.5% of blocks flip. 65.5% still pursue the $s = -1$ objective, the repair block-diagonalisation was designed to provide

Picard Contraction at Network Scale

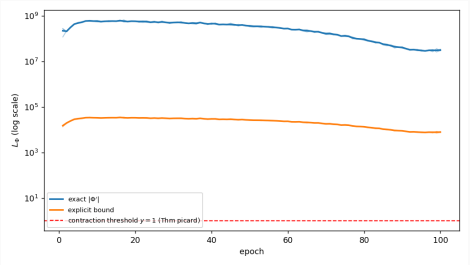


Figure. $|\Phi'|$ *and* L_Φ, `uCBOpt-iso` ($s = +1$).

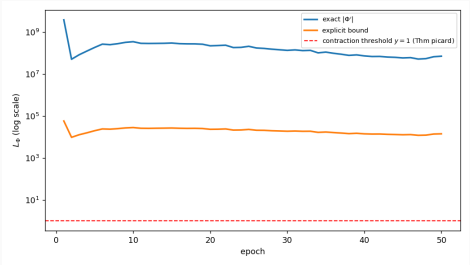


Figure. $|\Phi'|$ *and* L_Φ, `lCBOpt-iso` ($s = -1$).

- **Magnitude**: $|\Phi'|$ and L_Φ both exceed 1 from epoch 1, reaching 3×10^7 ($s = +1$) and 7×10^7 ($s = -1$) by the end of training at $d = 44{,}426$. $L_\Phi = 2d\|g\|^2/(u_{\min}(\eta)^4 a_{\min}^3)$, $g := \nabla\hat{\ell}(\eta)$, $a_{\min} := \epsilon_{\mathrm{rel}} \min_j(h_j + \delta)$, $u_{\min}(\eta) := \min_{\mathcal{I}_s} \|u_s\| > 0$
- **Caveat**: the convergence theorem's sufficient condition fails. But ϑ sits at the cap's boundary rather than converging to an interior fixed point, so it never tests the regime the theorem certifies

SAM Application and Sensitivity Results

The identity $\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\mathcal{E}_{\mathrm{SAM}}(\eta;\rho,0)$ is exact, checked on a trained checkpoint. The gap grows as $O(\rho^2)$, as expected from the first-order approximation $\varepsilon^{\star}_{\mathrm{SAM}}$: 1.79×10^{-3} at $\rho = 0.02$, up to 1.12×10^{-1} at $\rho = 0.1$.

$\pm 5\%$ sweep of r ($r_{\mathrm{nom}} = 0.50$) and ρ ($\bar{\rho}_{\mathrm{nom}} = 2.2165$, learned radius's mean), one seed, 50 epochs.

	$\times 0.95$	$\times 1.0$	$\times 1.05$
`lCBOpt-iso` (r)	0.8865	0.8876	0.8904
standard SAM [13] (ρ)	0.1373	0.2168	0.2399

Table. Accuracy under a $\pm 5\%$ sweep, learned versus standard SAM radius.

- **Sensitivity**: $\Delta_{\mathrm{adapt}} = 0.0039$ vs $\Delta_{\mathrm{SAM}} = 0.1026$, $26\times$ higher sensitivity for standard SAM
- **Radius scale**: $\bar{\rho}_{\mathrm{nom}} \approx 2.2$ is an order of magnitude above the SAM literature's typical $\rho \in [0.01, 0.2]$ [13]
- **Caveat**: ϑ saturates the anti-collapse bound rather than the interior optimum, so both are compared in suboptimal regimes

Discussion and Conclusion

Four Operational Limitations

1. **Safeguard infeasible**: the $s = -1$ safeguard fails on 100% of steps for `lCBOpt-LRD`. `lCBOpt-BD` partially repairs this (65.5% of blocks), but `uCBOpt-BD` remains numerically fragile (one benchmark seed diverged)
2. **Precision stays at the boundary**: `uCBOpt-iso` and `uCBOpt-diag` never reach the interior fixed point, sitting at the cap throughout training

   $\downarrow$
3. **Picard contraction fails**: $L_\Phi \sim 10^7$ to 10^8 at network scale for `uCBOpt-iso` and `uCBOpt-diag`
4. **SAM radius too large**: $\bar{\rho} \approx 2.2$ from `lCBOpt-iso` vs $\rho \in [0.01, 0.2]$ in the literature [13]

Conclusion and Open Questions

Contributions

1. Reproduced Singh et al. [1] within two standard deviations
2. Diagnosed misspecification, not divergence, as the source of candidate-target gaps
3. Learned the precision via a log-volume penalty: damped Picard fixed points for iso/diag, regularised closed-form safeguards for LRD/BD
4. Connected the lower CBO to SAM [13] and derived an adaptive perturbation radius
5. Assembled eight practical optimisers with a complexity discussion

Open questions

Multimodal/non-parametric candidates, annealing p, volume-penalty guarantees beyond the quadratic case, optimality of the LRD regulariser, joint (η, ϑ) convergence [18], whether the Gaussian relaxation for the learned SAM radius is theoretically sound.

Thank you

Questions?

Code Repositories

Three repositories accompany the report, one per contribution:

- **Reproduction** (Section: reproducing `uCBOpt` at fixed precision)
  `https://github.com/mariadass-gabriel/MDV_reprod`
- **Landscapes** (Section: landscapes on tractable posteriors)
  `https://github.com/mariadass-gabriel/MDV_lands`
- **Learned precision** (Sections: constructing and applying the learned precision)
  `https://github.com/mariadass-gabriel/MDV_spread`

Each repository contains the code and logs behind every number reported.

References

[1] Jasraj Singh et al. Maxitive Donsker–Varadhan formulation for possibilistic variational inference. *arXiv preprint arXiv:2511.21223*, 2026.

[2] Michael Scott Balch et al. Satellite conjunction analysis and the false confidence theorem. *Proceedings of the Royal Society A*, 2019.

[3] Lotfi A. Zadeh. Fuzzy sets as a basis for a theory of possibility. *Fuzzy Sets and Systems*, 1978.

[4] Didier Dubois et al. *Possibility Theory: An Approach to Computerized Processing of Uncertainty*. Plenum Press, 1988.

[5] Nong Minh Hieu et al. Decoupling epistemic and aleatoric uncertainties with possibility theory. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.

[6] Monroe D. Donsker et al. Asymptotic evaluation of certain Markov process expectations for large time, IV. *Communications on Pure and Applied Mathematics*, 1983.

References ii

[7] Thomas Laurent et al. The multilinear structure of ReLU networks. In *International Conference on Machine Learning (ICML)*, 2018.

[8] W. R. Mann. Mean value methods in iteration. *Proceedings of the American Mathematical Society*, 1953.

[9] Minh-Ngoc Tran et al. Variational Bayes on manifolds. *Statistics and Computing*, 2021.

[10] Aaron Mishkin et al. SLANG: Fast structured covariance approximations for Bayesian deep learning with natural gradient. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[11] Kenneth Rose. Deterministic annealing for clustering, compression, classification, regression, and related optimization problems. *Proceedings of the IEEE*, 1998.

[12] Shelvia Wongso. Lower CBO and SAM connection. Research note, HackMD, 2026.

[13] Pierre Foret et al. Sharpness-aware minimization for efficiently improving generalization. In *International Conference on Learning Representations (ICLR)*, 2021.

References iii

[14] Thomas Möllenhoff et al. SAM as an optimal relaxation of Bayes. In *International Conference on Learning Representations (ICLR)*, 2023.

[15] Yuesong Shen et al. Variational learning is effective for large deep networks. In *International Conference on Machine Learning (ICML)*, 2024.

[16] Ilya Loshchilov et al. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[17] Alexander D'Amour et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 2022.

[18] Vivek S. Borkar. *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press, 2008.